# Improving the Business Processes Management from the Knowledge Management

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## ABSTRACT
This paper proposes a model that integrates the knowledge management to the business processes management in order to improve the process performance. The proposal defines the knowledge management flow as an element that interacts with part workflows and provides rules and actions in the BPMN elements in order to improve the BPM. To achieve the aim, first, a survey was made in forty Colombian' companies in order to identify the state of the practice about business process management, knowledge management and their use through the workflows; then, different elements were identified and characterized in order to achieve the integration model proposed. This was experimented in forty business workflows of the companies analyzed. A use case shows how the interactions among model elements are, and how these can improve the process performance. This experience allowed to measuring the effect that the integration had in the business process management by means of a new metric that involve the knowledge management in the workflow. Finally, it was possible conclude that the knowledge management must be part to the BPM impacting directly the workflow, in order to improve the result offered to the organizational actors and customers.

## Keywords
Knowledge Management, Business Process Management, Workflow, Knowledge Service, Unstructured Information Service, Integration BPM and KM.

## 1. INTRODUCTION
Every day the information and knowledge management take more force in the organizations. Factors such as globalization, innovation, and sustainability in highly competitive markets accelerate the shift towards new organizational schemes. The knowledge management (KM) enables organizations to see the collective knowledge as a base element of innovation. This is possible through information and knowledge tools that provide environments to share the knowledge among employees and cooperative partners [1].


KMO '16, July 25-28, 2016, Hagen, Germany

DOI: http://dx.doi.org/10.1145/2925995.2925998

The knowledge management is a challenge for current organizations in order to be competitive and effective.

This is related to the organizations' strategies and how these can be used to take advantage adopting and mixing components of disciplines such as: business process management (BPM), innovated management styles, among others [2], [3]. In order to accomplish that, most organizations purchase large and expensive information systems for the functionality of their business units; also, they try of maintain big applications to support collaborative environments with the content management systems. This can be good and necessary. However, in the reality this is not easy to implement and integrate with the strategy. Some reasons might be because the companies have big problems of management and leadership, the strategy is developed in a functional organization but do not work by processes, or simply they do not recognize how to get return of investment from the knowledge management.

One of the aims of the BPM is to improve the performance of processes and provide good services to the customers. Although several proposals and efforts have been done, few of them have been treated the knowledge management how an important element to achieve it. This paper proposes a way to integrate the knowledge management to the organizational strategy through the business processes management. The proposal identifies knowledge flows as part of business workflow and provides rules and actions in the BPMN elements and standard measure elements in order to improve the BPM. To achieve the aim, a survey was made in three types of companies in order to identify the state of the practice about business process management, and the knowledge management; this allowed finding issues and elements that then were characterized in order to achieve the integration proposal. The proposal was experimented in one of the companies by means of case study. This experience allowed to establishing integration elements and measuring in the business process management. Finally, the proposal allows conclude the knowledge management is a valuable discipline that must be defined as part of the workflow in order to improve the result offered to the actors and customers.

This paper is ordered as following. Section 1, a background presents the main concepts used in the proposal development. Section 2, the research methodology is described. Section 3, shows the design of proposal. Section 4, the experience applying the proposal in a company is presented. Finally, the conclusions and future works are shown.

## 2. Literature Review

### 2.1 Business Process Management
The business process management (BPM) is an evolved management discipline. It is defined by a lifecycle which is composed by a set of stages such as Design & Analysis,

Configuration, Enactment, and Evaluation [4], or stages such as: Define, Model, Execute, Monitor, and Analyze [5].

BPM defines a business process as a set of activities or tasks and events jointed through workflows [6], [7]. It is identified as business unit that requires and provides information and knowledge services for the internal client as well as to the customers.

Burlton presents a model which involves information technology and knowledge management as elements that support the business process management in order to maintain a good relationship with customers and other stakeholders [7]. Some authors have regarded that a business process provides a way of "collaboration between organizational actors that are contributing to (or constraining) the satisfying of business objectives" [8]; also, the new technologies are providing different viewpoints to integrate the information to the of the organization's development [9]. The BPM evolve through the time and it is determined by several trends, methods and tools, therefore the organizations must consider that their employees and manager grow with a BMP-oriented organization [10]. Thus, the workflow management is the way process-aware software solutions and, in particular, in dedicated BPM-systems [11].

The BPM is supported by tools that provide automatized options to measure from different parameters. For instance, ARIS with the Process Performance Management provide a procedure "to collect process-relevant data from the IT systems, reconstruct process automatically and calculate key performance indicators for these processes." [12].

### *2.1.1 BPM Measures*

The business processes are measured because the organization needs to know if it is well defined through its processes, products or services in order to accomplish its strategy goals. Therefore, performance measure is related with process optimization increasing efficiency and effectiveness of the process, product or service [13]. Valis describes a measure model which defines what measure elements must be related with the task or activity in each processes or sub-process. The following measures are defined: Total Time (Start Time to End Time), Process Time ($\sum$ Process Time(Task)), Cost ($\sum$ Cost(Task)), and the measures by task such as: Process Time, and Cost (which is determined by the of task type in a specific business unit) [14]. Rosemann & Brocke propose a framework with six holistic measure elements: strategic alignment, governance, methods, information technology, people, and culture which provide the way to implement the BPM in practice [15], [16].

### *2.1.2 Business Workflow*

Georgakopoulos, Hornick, & Sheth define a workflow as "concept closely related to reengineering and automating business and information processes in an organization. A workflow may describe business process tasks at a conceptual level necessary for understanding, evaluating, and redesigning the business process." [17]. More recently, Van der Lans defines a workflow is the step by step where the tasks or activities of a business process in order to stablish a logical way that provides business services to different actors of the organizations. Different resources support the workflow definition; among these are the information systems and the data that flow through them" [18].

An important aspect of the workflow management is the modeling. Following international standards, the workflows are commonly modeled in the Business Process Model Notation (BPMN) with model elements such as task, event, gateway and flow [19].

### *2.1.3 The Business Process Maturity Model (BPMM)*

In this model the four level (White-box – statistically analyzed) and five level (White-box – statistical predictability), the processes are measured and controlled [20]. In order to achieve these levels, the organization defines or identifies the information flows in the process, every process element is standardized by means of the BPMN [19], and its subsequent implementation is supported by IT business systems [21]. Also, the workflows can be automated to facilitate its analysis and optimization by means of process model attributes like cost, and duration or error probability [22]. For instance, COBIT, ITIL, and CMM frameworks provide metrics based in the workflow information in order to add value for the customer [23].

## 2.2 Knowledge Management

The knowledge management (KM) is the way how the human resources or machine learning sharing and acquire experiences from different sources. In order to get value from the knowledge in the company it must be manage through of a lifecycle with the following stages: capture, store, transformation, transfer, and distribution [24]. From the perspective of [25] the KM lifecycle is saw as a process where each phase provide a set of activities that must be realized by the company in order to include the KM as part of its goals. They propose the following phases: Planning, Analyzing, Designing, Building, testing, and implementing, and Evaluating. Nissen refers lifecycle as "the kind of activity (e.g., creation, sharing, application) associated with knowledge flows. Flow time pertains to the length of time (e.g., minutes, days, years) required for knowledge to move from one person, organization, place, or time to another" [26].

Thus, the knowledge management lifecycle defined for the proposal is a set of adapted and standardized KMO stages that can be implemented at organizations.

*Create/acquire*. The knowledge is created by a human being when he/she has an experience that transforms his/her current state. Also, when the individual needs to improve his/her current knowledge, this can be acquired from different sources. Finally, this knowledge is registered in his/her brain like implicit knowledge. The knowledge acquisition also is associated to the machine learning where the computer can learn from declarative knowledge in order to improve to the user experiences [26]. The idea is that the implicit knowledge becomes explicit knowledge which will be recorded on the knowledge bases, and thus be used by others.

*Transfer*. The knowledge must be transferred to other individuals in order to achieve collaboration, cooperation or other actions that improve the life or the products or services into or intra organizations. In this stages as the below, it is very interesting to know about what, who, how, when, where and why the knowledge the knowledge must be used.

*Transform*. The individual accumulates experiences which can be combined with the acquired knowledge. Then, he/she can transform the current knowledge or produces new knowledge. This is an implicit action that is part of the above stages. A lot of individuals or machines learning transform the knowledge through the time in order to survey to different situations or improve those.

*Store/Retrieve*. Once the knowledge is created or transformed this must be recorded and store in computers in big knowledge repositories. Thus, the individuals can use mechanisms to retrieve knowledge according to the needs in different situations.

*Distribute*. The distribution of knowledge is a stage needed to complete the knowledge lifecycle. There are different ways, but the Web 2.0 is now the best way to achieve it.

### *2.2.1 SECI model*

Nonaka (1995) created the SECI model to manage the creation and transfer of knowledge into the organizations [27]. SECI defines an iterative cycle with four moments: socialization (tacit 2 tacit), externalization (tacit 2 explicit), combination (explicit 2 explicit), and internalization (explicit 2 tacit). Every time a cycle has been done different types of assets knowledge are created. The socialization mode suggest face to face communication, however it can include assets such as video conferencing, and virtual reality things like 360-degree video which will allow to recording social activities; in fact, the instant messages are considered as a formal medium to get information from business conversations [28]. The externalization mode includes assets such as know-how knowledge can be concreted form of information such text or symbols. The combination mode includes all form of knowledge recording mainly documents, reports, manuals, diagrams, databases, and information provided by knowledge techniques as lesson learned, successful cases, etc. The internalization mode can include assets such as organizational routines (business procedures), and the "know-how of daily operations embedded in actions and practices" [29].

Nonaka, Toyama, & Konno (2000) propose three types of knowledge assets: the experiential knowledge assets which involve knowledge acquired from images, symbols, and language; the routine knowledge assets which is created in the routine practices or techniques; and the systemic knowledge assets which have been stored and packaged as knowledge content [30]. Also, Tyagi, Cai, Yang, & Chambers (2015) identify knowledge assets such as know-how skills, patents, and technologies which might be used in different moments.

The relationship between knowledge lifecycle and SECI model is proposed by [26] through its knowledge flow model. That introduces two concepts "light mass" and "heavy mass". Tacit knowledge would be seen as "heavy mass" in the knowledge lifecycle, which means a "slow flow and a long flow time"; for instance, at activities as create/acquire knowledge at the socialization moment could regard it a "slow process" because the process might require several iterations or to be supported by other lifecycle activities in order to create the knowledge. On the other hand, the explicit knowledge would be seen as the "light mass", which means "rapid flows and short flow time"; for instance, at activities as transfer knowledge at the combination moment could regard it a "light process" because the knowledge has already been before and a team will use it in order to produce new knowledge.

## 3. Research Methodology

The methodology adopted in order to develop the proposal was the design science research [31]. This provides three cycles which establish a dynamics to identify the problem, know the state of the art, and does the proposal design. In a first cycle of Relevance, a survey in 40 Colombians' companies about the state of the practice in BPM, information management, and knowledge management was made. The companies were of three economic sectors: information technology (20 representative companies), environment agencies (5 representative companies), and higher education (5 representative companies). They had similar basic organizational characteristics such as: the companies are identified as learning companies, have between 350 and 2000 employees, the BPM is used by the company and the information and knowledge management is recognized. In the survey participated between the 30% and 50% of employees of each company on different organizational levels: strategic, management and operative.

The survey was divides in two sections: (1) BPM in the practice with questions about maturity level of business process, functional structure vs business process, human resources, BPM assessment method, indicators management, etc.; (2) Knowledge management in practice and its relation with the BPM. The questions were about maturity level of knowledge management, human capital capabilities to create, acquire, and transfer the knowledge during the process execution, and the social media used as part of the business process, etc.

The survey provided the following issues that must be attended by a model that involve the knowledge management as part of the business process management.

- Unawareness of knowledge management value of the most of members of the company. A lot of knowledge (tacit and explicit) is wasteful every day.
- Companies acquire expensive knowledge management tools in order to make something with the tacit and explicit knowledge. However the companies do not have the mechanisms or knowledge to design and implement the knowledge management into the organization.
- Companies even do not have the capacity or the knowledge to involve the knowledge management as part of the business process management.

This allowed to do in the Rigor cycle the state of the art where was identified in deep to have more clarity of the problem to resolve (see before section). Finally, in the design cycle the proposal was built and assessed (see following section).

## 4. Proposal Design

The proposal presents a model to involve the KM to the workflows. Figure 1 shows the set of process and knowledge assets that are part of the proposal. The elements are defined and designed in order to improve the business process performance.

The workflow performance is measured from the most used elements such as: people, cost, time, quality, technology, and information. In addition, the proposal provides the knowledge as an element that can be measured through the workflow.

## 4.1 The Workflow Components

The workflow management is determined by the BPM lifecycle. Every stage is a set of activities in order to every workflow accomplishes the strategy goals of the company.

### *4.1.1 BPMN Elements*

The workflow is designed in the BPMN (*Business Process Model Notation*) [33], and its elements are the basis for determining how the *Knowledge Management Flow* must be integrated to the workflow. Hence, the model will provide a meaning for the knowledge that is managed for each workflow element, and vice versa. Table 1 presents the meaning of the knowledge for every BPMN model element.

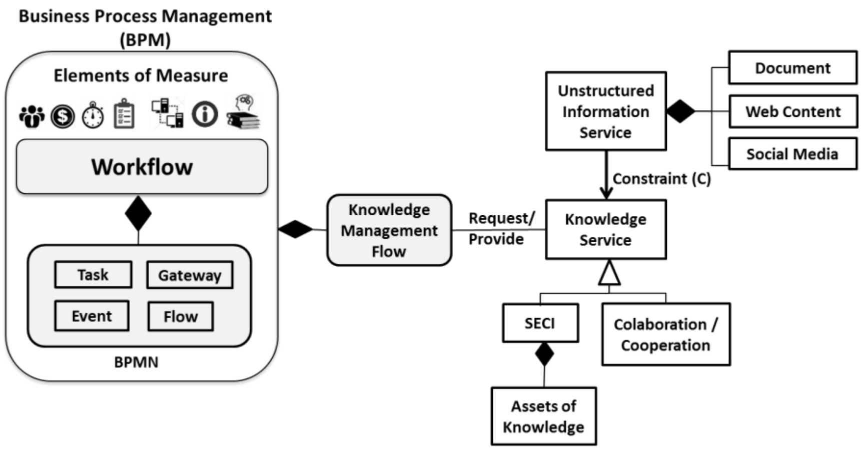


**Figure 1. Model reference to integrate the Knowledge Management Flow to the Business Workflow.**

Table 1. The meaning of Knowledge flow at the BPMN elements.

| BPMN model elements | **Knowledge meaning** |
|---|---|
| Task | Space where people (human talent) acquire, share or register knowledge |
| Event | Something happen that could trigger a task with specific knowledge. For example, when an employee is assigned to new task, a knowledge mode must be activated in order to receive a new knowledge |
| Gateway | Validation of a business rule which in turn will take actions, and then it might require or provide knowledge |
| Flow | Carries knowledge flowing through workflow |

### *4.1.2 Workflow Measure Elements*

One of the aims of the BPM is to provide the way to measure the organization through its processes, in order to improve its performance. Therefore it is necessary to define how it will be measured. The model provides seven workflow measure resources.

**Actors (people)** : This is the human resource assigned to every task (or process) in order to make a specific procedure. They are the main character for both the BPM activities and knowledge activities. They are measured by the compliance of the business activities and the knowledge management activities.

**Time** . This is the duration of a task to run. The sum of the duration of all process tasks, it is the workflow execution time. A task can require time to acquire or transfer knowledge, and not only to modify information in the information systems. The time as measure variable will depend of other resources in order to calculate its value.

Quality . This is resource applied to the tasks or processes in order to verify and validate their correct execution. The quality resource will depend from the compliance required for a task or process. This is defined before the process run. The knowledge management is a way to improve the quality activities.

Technology . This resource refers to the information technology or the infrastructure used by the workflow or task. This is used when it is necessary to measure by the cost of the technology used by the process. From the knowledge management, it is important to identify the use of technology like content management systems or knowledge repositories.

Information . This resource refers to the system information, relational databases, and unstructured information used by the workflow or task. This resource has a strong relationship with the knowledge because one of them can be source of the other.

Knowledge . This resource will be measured from the knowledge management flows (which will be explained in the development of this paper). These will provide the number of times that both the tacit or explicit knowledge have been used during the process run.

Cost . This resource is the summary of the cost of the before resources mentioned.

## 4.2 Knowledge Management Flows

A *Knowledge Management Flow* (KMF) is an instance of the knowledge management lifecycle which interacts directly with the workflows in runtime through of knowledge actions such as: Capture/Acquire, Transform, Transfer, Store/Retrieve and Distribution. Thus, a KMF provides knowledge management actions in order to the workflows improve the process performance.

Figure 2 shows how is integrated the KMF to the business workflow through of Knowledge Services (KS). These are the dynamics way to support the needs required by a specific workflow. Each KS is defined as a rule (or set of rules) that manage knowledge actions for tasks or Events defined by the workflow. In fact, each KS is defined or scheduled according to the needs of managing the knowledge for a specific workflow.

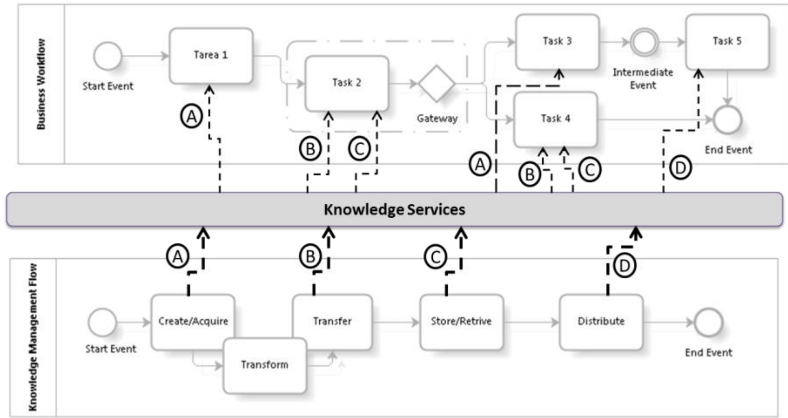


**Figure 2. Integrating Knowledge Services to the business workflow.**

The knowledge needs are classified as fixed or variable. A knowledge action is fixed on to the workflow when it is required permanently to improve the workflow performance. A knowledge action is variable for a workflow when it is required in a task for an unexpected event. This kind of events can affect the performance of workflow and not always are positive.

For instance, a case of use about fixed needs. The Task 1 is defined with a KS of acquisition (*A*) because people assigned to the task need it in order to execute the task. Thus, the use of knowledge in the task can be measured. By default, the knowledge is acquired and retained by each individual (tacit knowledge) and sometimes recorded into the organization (explicit knowledge). The other side, *Task 2* might require two KS: transfer (*B*) y store/retrieve (*C*), because in this task specifically the knowledge must be transferred to other actors. Here, the KS *B* provides rules or elements to control the transfer, and KS C is responsible for storing the transferred knowledge in order to others can collaborate or cooperate in next tasks. The *Task 4* requires/provides preserved digital documents or historical web content with knowledge that will be the base to distribute to others; therefore, two KS are required in this task: retrieve and distribute. It is needed to collaborate or cooperate with other actors or groups. So on, for each of the tasks should assess the need for services knowledge to improve process performance.

### *4.2.1 Knowledge Service Modes*

The KMF guarantees the creation and transfer of knowledge supported by the SECI model. Every KS must be defined in one or more modes provided for the SECI Model in order to have control and measure more objectively.

- *<Socialization Service – SS>*. This service is required or provided when actor resources of a specific task consider that a conversation must be information to support the workflow. If the service is required, this can be two way existing information or provided by a knowledge management flow when an informal or formal business conversation must be recorded or retrieved as part of a workflow task. The outcome is a conversation asset such as video conferencing, and virtual reality things like 360-degree video. In the service is us recorded in as a new knowledge asset that has a metadata related to a content management system.

The KS can be used when a User Task is executed or when a Message Flow is connected between a task and a customer pool or others kind of pools. BPMN elements: Manual Task (without social media records) or Web Service task when the individuals use social media assets.

- <*Externalization Service – ES*>. This service is required or provided by a KS in order to individuals of the workflow transfer knowledge to groups involved in different tasks. Assets like forums, lessons learned, best practices, slideshows, and documents that support the learning must be recorded. These asset help to improve the performance's workflow either in time or quality: Manual Task (without social media records) or Web Service task when the individuals use social media assets. An ES interact with an UIS in order to the unstructured data be treated in an information flow.
- <*Combination Service – CS*>. This service is required or provided by a KS when explicit knowledge is interchanged from different sources like e-books, digital reports or documents, spreadsheets, organizational memory, etc., in order to create new knowledge. The outcomes must be recorded in an organizational knowledge base composed mainly by unstructured information.
- <*Internalization Service – IS*>. This service is required or provided for a workflow when individuals or machines learning going to learn about explicit information recorded in the knowledge base. Different types of tasks can use this service.

### *4.2.2 The Collaboration/Cooperation Services – CCS*

This kind of KS might be very common when a workflow is defined. By natural, the workflow actors work collaboratively; however, the knowledge produced during the collaboration is not recorded in a knowledge base. Thus, the KS required or provided by a workflow when individuals or groups share knowledge across the workflow. This service gets information or knowledge through the collaborative tools like wiki, blogs, etc.

### *4.2.3 Unstructured Information Service (UIS)*

The explicit knowledge can be recorded in documents, web content systems or different type of social media like wikis or blogs. This is called unstructured information because they do not have a defined structured in order to be store in a database. However, not all unstructured information is knowledge. Thus, the proposal only considers the use of UISs with knowledge stored in knowledge bases. A UIS provides the metadata and content to be registered as new knowledge. The knowledge base is increased every time that the documents, web content, or social media are used by actors in the workflow as part of internalization or combination modes.

All experiences acquired by the actors that can be sharing during the workflow must be recorded in the base of knowledge using knowledge techniques and tools (e.g. lesson learned).

# 5. Measuring the interaction between the KMF and Workflows

The knowledge management Flow provides different elements that can improve the process performance. In order to achieve that the proposal defines the following measuring elements that are part of the interaction between the KMF and the workflow.

- Technique of knowledge (ToK). This is the way how the organization carries out a particular knowledge activity during the knowledge lifecycle [32]. For example: lesson learned, brainstorming, etc.
- Knowledge Service (KS). This is the KS declared for the KMF in order to drive the use of knowledge during the workflow execution.
- Knowledge Tool (KT). This is something used in the performance of a knowledge activity. For example: flip chart, e-forms, video, etc. A KT allows that an individual or group used repeatedly a knowledge asset.

With these three measure elements, the metric of the interaction between a KMF and a workflow (Ik&w) is defined in order to improve the process performance. The metric must be applied when a KMF is defined for a specific workflow, in other words when BPMN element requires/provides knowledge service (KS). The metric is defined as follow.

Identifying the iteration between knowledge and the workflow:

Ik&w = BPMN Element *require/provide* KS *includes* (ToK+KT) (1)

Process Performance = Sum((a) + (b)) (2)

Ik&w * Actor (a) Ik&w * Time (b)

# 6. KMF in Practice

The proposal was tested through of three use cases, one per each type of companies involved in the survey. The proposal was applied and validated during four months to selected processes of the operational level. The test was carried out by a team of enterprise architects (business domain) and business people.

### *6.1.1 The Process*

In order to get the experience for every use case, the enterprise architects team follows the following method.

1. What elements are part of the workflow? The following steps were made: first, the workflow was understood from its services and its contribution to the business goals. Second, every BPMN workflow element: task, gateway, event and flow were understood from its functionality and relationship through the flow. Also, it were identified the information systems, documentation and governance which were supported the workflow or every task.
2. Who takes part at each workflow task? It is about its function at the task and the time that use to execute the task. As well as the tacit knowledge of every person, and the current knowledge techniques that they used in order to improve their work.
3. When the knowledge techniques must be used through the workflow?
4. How the workflow is improved using the knowledge management techniques?

### *6.1.2 The Use Case*

Figure 3 show the interaction between KMF and the process of *Software Requirement Order* from the information technology companies. The process offers the service of *Delivery Software by Demand* in order to attend the IT requirements from different customers. The process involves two actors: the IT Support area, and the Customer. The IT Support is an autonomous area into the company and leads the process. Therefore, it can invoice services and software packages. The Customer is the administrator of an information system who is inside of the company and knows how a why a Software Requirement Order must be managed. A Customer fills a web form to create a software requirement order with the following data: identification, date, requirement and constraints. Once this happens, the IT Support area <Analyze the Order> and makes the process to deliver the software that by demand has been required. Figure 3 show the interaction between KMF and the process of Software Requirement Order.

The information is captured in a document by the <Create requirement order> task requires a UIS in order to store it in the knowledge base. Thus, a KS is required to create/acquire a new knowledge that will serve as base of a CCS. Once the document can be retrieved for the <Analyze Order> task this can be part of other KSs like to transfer to other team with additional knowledge for the taken decisions. Thus, the flow is activated in order to know lessons learned from previous workflow run then another <Knowledge Service> is required.

Once the decision is made tasks as <Create the software Package> shall include the SECI services in order to combine knowledge has been created from the previous experiences to run the other tasks. Also, the <*intermediate event*> can lead a meeting of group of experts before or during the execution of that task, or could require a <*Collaborative/Cooperative Service*> to activate an action to meet the participants. This task should be improved in cost or time. Likewise, the <*Send software package*> task may be supplemented with a knowledge distribution service through the different web services such as a website (intranet or portal), a wiki, a blog, or other social media services.

# 7. Results and Discussion

In order to achieve the experience to incorporating the KMF proposal, the team tested the model running it through several simulations with a set of values over the workflow. A. “iteration” was defined as a set of simulation runs using some different knowledge techniques that were integrated to the workflow model elements. Table 2 present some iteration run in a period of time. The experience allows identifying how many times must be invested by an individual or a team in order to improve the workflow performance.

# 8. Conclusions and Future Works

The proposal described in the paper provides an interaction between BPM and Knowledge Management. A model with different stages of the knowledge management is defined in order to improve the process performance.

The companies must make questions about what are the knowledge sources (customers, suppliers, collaborators, business areas, etc.) and why they change? How much is the cost of the knowledge management for each business unit? Why the knowledge must be preserved and careful? Why the knowledge are essential to survive? How can be competitive with the knowledge required or provided by the company? Are the customers happy with the knowledge transferred?

Consequently, the organization must draw upon value resources as the knowledge processed inside and outside of the organization. However, to find the way to integrate these big aspects into the business process management style it is not an easy task. This paper proposes a way to integrate the knowledge management to the business processes management.

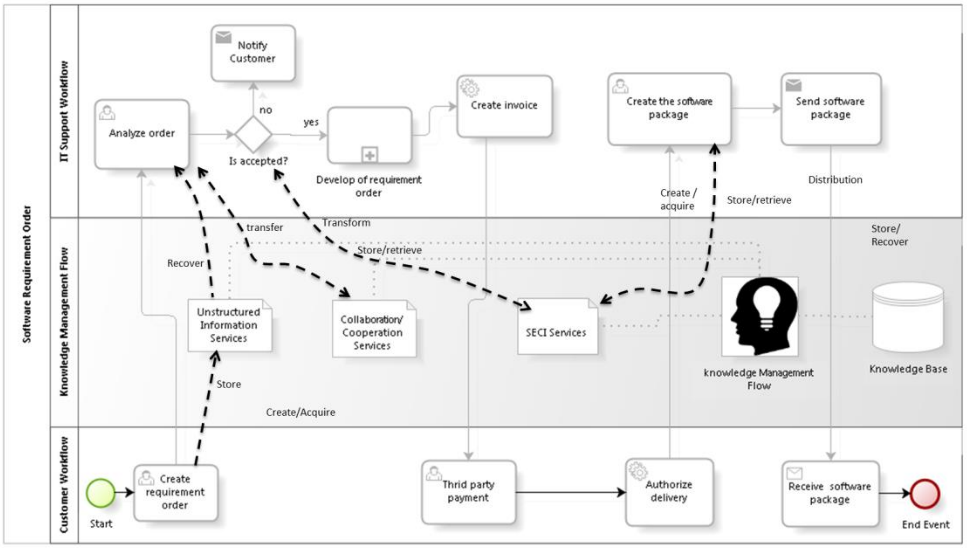


**Figure 3. KMF applied to the use case of the workflow of "Software Requirement Order".**

**Table 2. KMF Model Simulation**

| Iter. | Workflow Element | Order Characteristics | Knowledge activity | Human Resources | Time | Reduced time |
|---|---|---|---|---|---|---|
| 1 | Task: Analize the orden | Requirement about supplier information | Nothing | Analyst | 1h | 0 |
| 1 | Task: Analize the orden | Requirement about supplier information | Nothing | Analysts team | 1:10h | -10m |
| 2 | Task: Analize the orden | Requirement about supplier information | Create Lesson learned | Analyst | 1h | 0 |
| 2 | Task: Analize the orden | customer needs | Recover Lesson Learned | Analyst | 30m | 30m |

The knowledge management is necessary as a complement to implement all the steps defined in the workflows to improve the business process consistency, coherence and security and then improve the process performance. The different types of tasks, events and gateways that provides BPMN identifies criteria for designing the workflows and knowledge of every process in the organization.

In order to define the needs of knowledge management for workflows, the BPM in the organization must be very well defined and to have aware that the knowledge management might improve the performance. Knowledge management depends on a large percentage of human talent. Therefore the tasks of creating and transferring knowledge during the execution of a business process should be predetermined and in the workflow to achieve the dynamics of collective intelligence in the organization.

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